restricted by political and ethical debates over autonomous lethal operations and civilian collateral risks. Logistical support – such as the need for dedicated launch and recovery infrastructure, high-bandwidth communication networks, and specialized maintenance teams also limited their deployment flexibility.

4.6.4 Cost and accessibility issues

Although UAVs like the Wing Loong I introduced cost-effective solutions, most third-generation platforms remained expensive to procure and operate, particularly for smaller nations. The integration of high-performance engines, advanced sensors, and stealth materials increased lifecycle costs, while supply chain dependencies restricted local manufacturing capabilities. Moreover, limited export authorization from major developers (such as the United States and Israel) constrained global accessibility, creating a gap between technological advancement and widespread adoption.

4.6.5 Ethical, legal, and regulatory challenges

The deployment of armed drones introduced complex ethical and legal dilemmas, particularly in relation to autonomous decision-making and targeted strikes. Questions regarding accountability, rules of engagement, and compliance with international humanitarian law became central to UAV operations. Many nations struggled to develop comprehensive airspace integration policies and autonomy regulations, leading to fragmented global standards.

4.7 Case Study: Operational Deployment and Impact of Second-generation Drones

The third generation of drones marked a transformative phase in unmanned aviation, characterized by the large-scale operational deployment of long-